

20/1/25
Monday

Module - 2

Eulerian & Hamiltonian Graphs

Euler graph:

A graph which contains an euler line is called an euler graph.

Euler line:

A closed walk which should pass through all the edges of G exactly once.

Euler Path:

It is a path that passes through all the edges of the graph at most once i.e. starting & ending vertices are different & no vertex appears more than once.

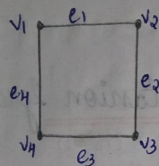
Euler circuit:

It is a circuit which passes through all the edges of the graph at most once. Here ~~or~~ no vertices will appear more than once except starting & ending vertices.

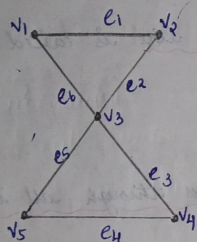
Points to remember for euler line:

- closed walk
- Passes through all the edges of the graph exactly once.
- vertices can be repeated.

Eg:



Clearly the graph is an Euler graph since it contains an Euler line
 $v_1 e_1 v_2 e_2 v_3 e_3 v_4 e_4 v_1$.



Clearly the graph is an Euler graph since it contains an Euler line

$v_1 e_1 v_2 e_2 v_3 e_3 v_4 e_4 v_5 e_5 v_3 e_6 v_1$

But it is ~~not~~ an Euler circuit, since the vertex v_3 comes twice.

Hamiltonian graph:

→ Hamiltonian path:

A path consisting of all the vertices of a graph is called the Hamiltonian path.

→ Hamiltonian circuit:

A closed walk passes through every vertex exactly once except the starting & ending vertices.

→ Hamiltonian graph:

A circuit in a connected graph is Hamiltonian if it includes all the vertices of G exactly once.

Note:

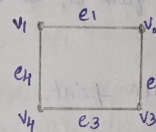
→ Let G be a graph with ' n ' vertices & G is Hamiltonian graph then, it has exactly ' n ' edges.

→ If we remove any one edge from a Hamiltonian circuit, we are left with Hamiltonian path.

→ Length of the Hamiltonian path in a connected graph of ' n ' vertices is always ' $n-1$ '.

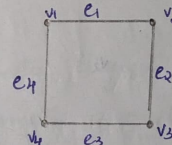
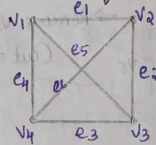
A) Draw a graph which is:

a) Euler & Hamiltonian graph:



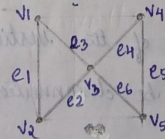
$v_1 e_1 v_2 e_2 v_3 e_3 v_4 e_4 v_1$ → Euler line

b) Not Euler graph but Hamiltonian graph:



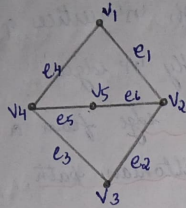
→ circuit.

c) Euler but not Hamiltonian graph:



$v_1 e_1 v_2 e_2 v_3 e_3 v_4 e_4 v_5 e_5 v_1$ → Euler line

4) Not euler not hamiltonian:



→ Cut vertex:

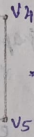
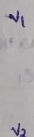
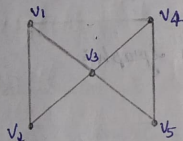
A cut vertex of a graph $G=(V,E)$ is a vertex V such that $G-V$ has more connected components than G .

OR

When we are deleting the vertex V from G , it will become disconnected.

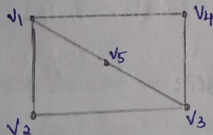
Cut vertex is also called articulation point.

Eg: ①



⇒ Remove v_3
(cut vertex)

②



} cut vertex
absent

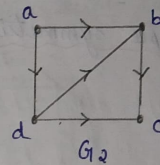
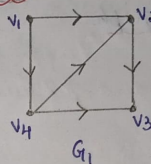
When we are removing any one of the vertices v_1, v_2, v_3, v_4 & v_5 , the graph will not be disconnected. So, there is no cut vertex.

Since there is no cut vertex in the above graph, the graph is not hamiltonian. (One vertex will be repeated).

→ Isomorphism diagram:

For two graphs to be isomorphic not only must their corresponding undirected graphs be isomorphic, but also the directions of the corresponding edges must also agree.

Eg:



Here G_1 & G_2 are isomorphic.

Types of Diagram:

→ Simple diagram

→ Symmetric diagram

→ Asymmetric diagram

→ Complete diagram

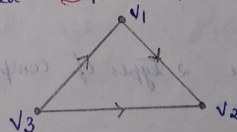
(i) complete symmetric diagram

(ii) complete asymmetric diagram

→ Pseudo symmetric diagram.

1. Simple diagram:

A diagram that has no self loop or parallel edges, are called simple diagram.

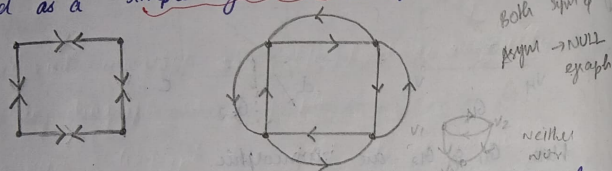


2) Symmetric diagram:

The diagrams in which for every edge (a, b) (ie from a to b) then there is also an edge (b, a) (ie from b to a).

* A diagram that is both simple & symmetric is called as a simple symmetric diagram.

Eg:

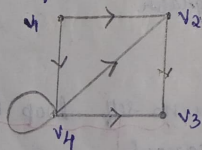


Both the graphs are simple symmetric & symmetric also.

3) Asymmetric diagram:

* Diagrams that have atmost one directed edge b/w a pair of vertices, but are allowed to have self loop are called asymmetric or anti-symmetric diagrams.

A diagram that is both simple & asymmetric is simple asymmetric diagram.



4) Complete diagrams:

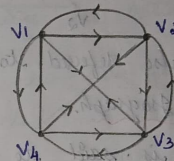
For diagrams we have 2 types of complete diagrams:

1. Complete symmetric diagrams
2. Complete asymmetric diagrams.

1. Complete symmetric diagrams:

It is a complete diagram in which there is exactly one directed edge from every vertex to every other vertices.

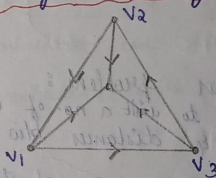
Eg:



2. Complete asymmetric diagrams:

It is an asymmetric diagram in which there are exactly one edge b/w every pair of vertices.

Eg:



A complete asymmetric diagram of ' n ' vertices contains $\frac{n(n-1)}{2}$ edges.

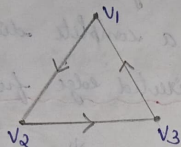
But a complete symmetric diagram of ' n ' vertices contains $n(n-1)$ edges.

A complete asymmetric diagram is also called a tournament or complex tournament.

5) Pseudo-symmetric diagraphs:

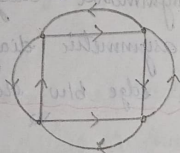
A diagraph is said to be balanced if for every vertex v_i , the indegree equals the outdegree.

i.e. $d^+(v_i) = d^-(v_i)$



A balanced diagraph also referred to as a pseudo-symmetric diagraph or isograph.

A balanced diagraph is said to be regular if every vertex has same indegree & outdegree as every other vertex.



21/1/26
Tuesday

Travelling Salesman problem:
A salesman is required to visit a no. of cities during a trip. Given a list of cities & distances b/w each pair of cities, we have to find the shortest possible route that visits each city exactly once & returns to the origin city.

The TSP is a problem in Graph theory requiring the most efficient (i.e. the least total distance) hamiltonian cycle a salesman can walk through each of n cities.

TSP can be modelled as an undirected graph (undirected weighted graph) such that the cities are graph's vertices, paths are graph's edges & path distance is edge's weight. The graph is a complete graph. If no path exist b/w two cities, adding an arbitrary long edge will complete the graph, without affecting the optimal TSP tour.

In TSP, we have to find the minimum weight hamiltonian cycle. total travelling

Algorithm:

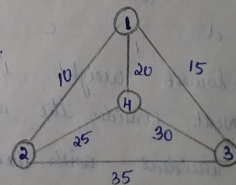
1. Consider city 1 as the starting & ending points.
2. Generate all $(n-1)!$ factorial permutations of the cities.
3. Calculate the cost of every permutation & keep track of minimum cost permutations.
4. Return permutation with minimum cost.

NOTE:

If we try to determine the solution of the problem, we could end up with $(n-1)!$ possible solutions.

Eg: If there are 7 cities we have $6!$ different ways to travel.

Fix TSP of the following graph.



In the above figure $n=4$, the set of all permutations of the cities & their cost are given below. Here we must have $(4-1)!$ i.e. $3! = 6$ solutions.

(a) $1-2-4-3-1$

$$10 + 25 + 30 + 15 = 80$$

(b) $1-4-2-3-1$

$$20 + 25 + 35 + 15 = 95$$

(c) $1-3-4-2-1$

$$15 + 30 + 25 + 10 = 80$$

(d) $1-3-2-4-1$

$$15 + 35 + 25 + 20 = 95$$

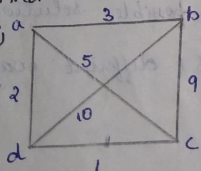
(e) $1-2-3-4-1$

$$10 + 35 + 30 + 20 = 95$$

(f) $1-4-3-2-1$

$$20 + 30 + 35 + 10 = 95$$

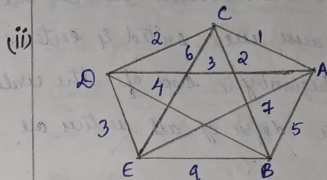
24/10 Friday Find the solutions for following TSP.



Ans: Consider an arbitrary vertex C . The edges incident with C with the lowest weight i.e. the edge CD , with weight one. Next consider the edge with the lowest weight incident with HD , is the edge DA , & continue the

process, we obtain the hamiltonian cycle

$$C-d-a-b-c = 1 + 2 + 3 + 9 = 15$$



Ans: Clear. Consider the arbitrary vertex C or A with lowest weight i.e. the edge CA , with weight one. Next consider another edge with lowest weight incident to C or A . & continue the process we get different hamiltonian cycles.

1. $C-A-B-D-E-C$
 $1 + 5 + 4 + 3 + 6 = 19$

2. $C-A-D-E-B-C$
 $1 + 3 + 3 + 9 + 2 = 18$

4. $A-C-B-D-E-A$
 $1 + 2 + 4 + 3 + 7 = 17$

3. $A-C-D-E-B-A$
 $1 + 2 + 3 + 9 + 5 = 20$

Theorem - 1:

A connected graph G is an euler graph if & only if all the vertices of G are of even degree.

Proof:

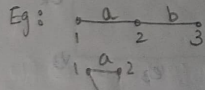
Suppose that G is an euler graph i.e. it contains an euler line. Since euler line is a closed walk while

tracing the walk, it is observed that every time the walk entered and with the other exited. That is true not only for intermediate vertices, but also for terminal vertices, because we exited & entered the same vertex at the beginning & end of the walk respectively. This shows that the degree of all vertices are same.

Conversely assume that all vertices of G are of even degree. Now we construct a walk starting at an arbitrary vertex v & going through the edge of G and no edge is traced more than once. We continue tracing as far as possible, since every vertex is of even degree, we can exit from everywhere we enter, the tracing cannot stop at any vertex but v . & since v is also of even degree, we shall eventually reach v when the tracing comes to an end. If this closed walk h , we just traced includes all edges of G , G is an euler graph.

If not, we remove from G all the edges in h & obtain a subgraph h_1 of G formed by the remaining edges. Since both G & h have all their vertices of even degree, the degree of this vertex of h_1 are also even. Moreover h_1 must touch ^{at least} at one vertex, ^{but} because G is connected. Starting from a , we can again construct a new walk in graph h_1 , since all vertices of h_1 are of even degree this walk in h_1

must terminate at vertex a , but this walk in h_1 can be combined with h to form a new walk which starts & end at vertex v & has more edges than h . This process can be repeated until we obtain a closed walk then ^{that} traverses all the edges of G . Thus G is an euler graph. ^{G contains} euler tour.



Operations on graphs:

→ Union, Intersection & ring sum of 2 graphs.

1. Union:

Let $G_1(V_1, E_1)$ & $G_2(V_2, E_2)$ be 2 graphs, then the union of G_1 & G_2 is denoted by $G_1 \cup G_2$ & is defined by

$$G_1 \cup G_2 = G_3(V, E) \text{ where } V = V_1 \cup V_2 \text{ & } E = E_1 \cup E_2$$

2. Intersection:

Intersection of the graphs G_1 & G_2 is denoted by $G_1 \cap G_2$ & is defined by:

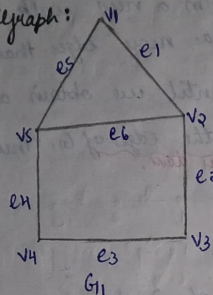
$$G_1 \cap G_2 = G_4(V, E) \text{ where } V = V_1 \cap V_2 \text{ & } E = E_1 \cap E_2$$

3. Ring sum:

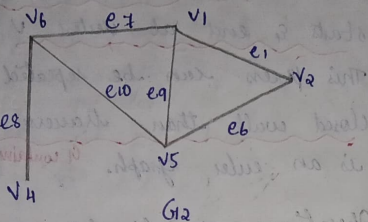
The ring sum of the 2 graphs G_1 & G_2 is denoted by $G_1 \oplus G_2$, & is defined by:

$$G_1 \oplus G_2 = G_5(V, E) \text{ where } V = V_1 \cup V_2 \text{ & } E = E_1 \cup E_2 - E_1 \cap E_2$$

22/1/25 Monday Find union, intersection & ring sum of the following graph:



G_1



G_2

Ans: $G_1(V_1, E_1)$

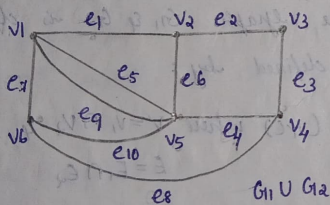
$$V_1 = \{v_1, v_2, v_3, v_4, v_5\}$$

$$E_1 = \{e_1, e_2, e_3, e_4, e_5, e_6\}$$

union: $G_1 \cup G_2$

$$V = V_1 \cup V_2 = \{v_1, v_2, v_3, v_4, v_5, v_6\}$$

$$E = E_1 \cup E_2 = \{e_1, e_2, e_3, e_4, e_5, e_6, e_7, e_8, e_9, e_{10}\}$$

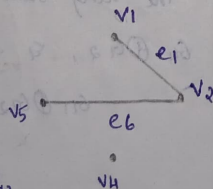


$G_1 \cup G_2$

Intersection: $G_1 \cap G_2$

$$V_E = V_1 \cap V_2 = \{v_1, v_2, v_4, v_5\}$$

$$V = E_1 \cap E_2 = \{e_1, e_6\}$$

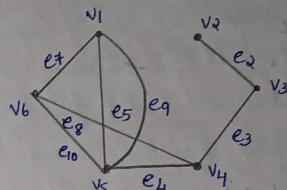


$G_1 \cap G_2$

Ring sum: $G_1 \oplus G_2$

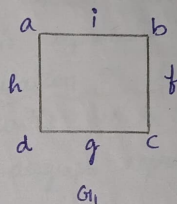
$$V = V_1 \cup V_2 = \{v_1, v_2, v_3, v_4, v_5, v_6\}$$

$$E = E_1 \cup E_2 - E_1 \cap E_2 = \{e_2, e_3, e_4, e_5, e_7, e_8, e_9, e_{10}\}$$

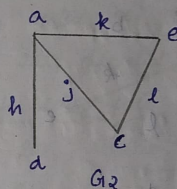


$G_1 \oplus G_2$

Q)



G_1



G_2

Ans: $G_1(V_1, E_1)$

$$V_1 = \{a, b, c, d\}$$

$$E_1 = \{i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z\}$$

$G_2(V_2, E_2)$

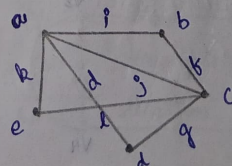
$$V_2 = \{a, b, c, d\}$$

$$E_2 = \{k, l, j, h\}$$

union: $G_1 \cup G_2$

$$V = V_1 \cup V_2 = \{a, b, c, d\}$$

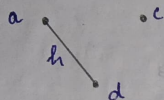
$$E = E_1 \cup E_2 = \{i, j, k, l, t, g, h\}$$



Intersection : $G_1 \cap G_2$

$$V = V_1 \cap V_2 = \{a, c, d\}$$

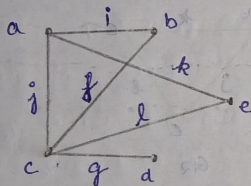
$$E = E_1 \cap E_2 = \{h\}$$



Ring sum : $G_1 \oplus G_2$

$$V = V_1 \cup V_2 = \{a, b, c, d, e\}$$

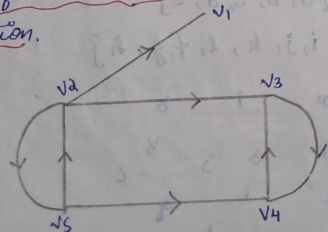
$$E = E_1 \cup E_2 - E_1 \cap E_2 = \{f, g, i, j, k, l\}$$



28/1/25
Tuesday

Directed graph:

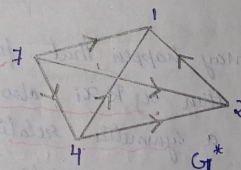
A directed graph (diagraph) G_1 , consists of set of vertices, $V = \{v_1, v_2, \dots\}$, a set of edges, $E = \{e_1, e_2, \dots\}$ & a mapping ϕ that maps every edge onto some ordered pair of vertices (v_i, v_j) i.e. each edge of G_1 has a direction.



Diagraph & binary relation:

In a set of objects X , where $X = \{x_1, x_2, \dots\}$ & binary relation R b/w the pairs (x_i, x_j) may exist in which case we write $x_i R x_j$, and say that x_i has relation R to x_j , these relations are defined on pairs of numbers of the set, and this is why the name binary relation.

A diagraph is the most natural way of representing a binary relation on set X . Each x_i element of X , is represented by a vertex x_i . If x_i has the specified relation R to x_j , the directed edge is drawn from vertex x_i to x_j , for every pair (x_i, x_j) . The following diagraph represents the relation, " x is greater than, y " on a set containing four numbers $\{1, 2, 4, 7\}$.



Diagraph of binary relation:

→ Every binary relation on a finite set can be represented by a diagraph without parallel edges. Conversely every diagraph without a parallel edges defines a binary relation on the set of vertices.

Reflexive relation:

→ For some relation R , it may happen that every element is in relation R to itself.

Eg: A number is always equal to itself, or a line is always parallel to itself.

→ Such a relation R on a set X that satisfies $x_i R x_i$ for every x_i element of X is called a reflexive relation.

→ The diagram of a reflexive relation will have a self loop at every vertex, such a diagram representing a reflexive binary relation on its vertex set may be called a reflexive diagram.

→ A diagram in which no vertex has a self loop is called an irreflexive graph.

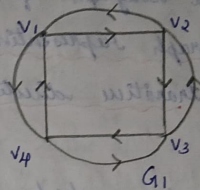
Symmetric relation:

→ For some relation R , it may happen that for all x_i & x_j if $x_i R x_j$ holds, then $x_j R x_i$ also holds. Such a relation is called a symmetric relation.

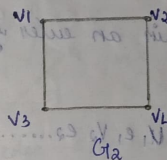
→ Eg: $x_i = x_j$ is both symmetric & reflexive relation.
"is spouse of" is a symmetric relation but not reflexive.

→ The diagram of a symmetric relation is a symmetric diagram, because for every directed edge from vertex x_i to x_j , there is a directed edge from x_j to x_i .

→ Eg:



G_1 is a graph which is symmetric but not reflexive. The same above relation can be represented by drawing just one undirected edge b/w every pair of vertices that are related as in below graph G_2 .



Note:

→ Every undirected graph is a representation of some symmetric binary relation.
→ Every undirected graph with e edges can be thought of as a symmetric diagram with $2e$ directed edges.

Transitive relation:

A relation R is said to be transitive if for any 3 elements x_i, x_j, x_k in the set $x_i R x_j$ & $x_j R x_k$ always implies $x_i R x_k$.

Eg: A binary relation (is greater than) is transitive.
 $x_i > x_j, x_j > x_k \Rightarrow x_i > x_k$

Graph G^* is the diagram of a transitive binary relation. A diagraph representing a transitive relation is called transitive directed graph.

3/1/25
Fleury's Algorithm:

→ It describes a procedure which constructs an euler tour in an euler graph. The algorithm can be described as given below:

→ Steps - Fleury's algorithm:

1. Choose any vertex v_0 in an euler graph G , set $w_0 = \{v_0\}$.

2. If the trail $w_i = \{v_1 e_1 v_2 e_2 \dots e_{i-1} v_i\}$ has been chosen such that all edges are selected all different then choose another edge e_i such that e_i is different from all previously selected edges.

→ e_i is different from all previously selected edges.

→ e_i is incident with v_i .

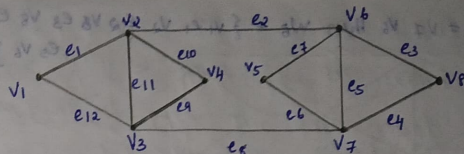
→ unless there is no alternative, e_i is not a cut edge of the edge deleted subgraph $G - \{e_1, e_2, \dots, e_{i-1}\}$.

3. If w_i covers every edge of G , stop the process.

Otherwise go to step 2.

Ex: Step by step illustration of Fleury's algorithm:

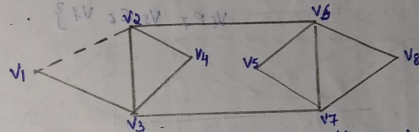
consider the following graph



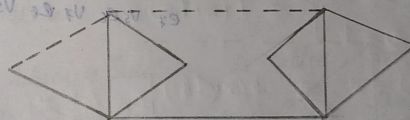
Ans: Since every vertex of G has been even degree, G is an euler graph. A step by step illustration of fleury's algorithm is given below:

1. Choose v_1 , arbitrary vertex & $w_1 = \{v_1\}$.

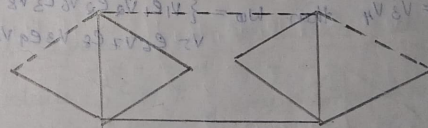
1.2. Choose an edge $e_1 = v_1 v_2$ then $w_2 = \{v_1 e_1 v_2\}$ & we have



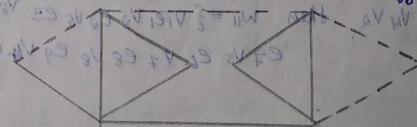
2. Choose the edge $e_2 = v_2 v_6$ then $w_3 = \{v_1 e_1 v_2 e_2 v_6\}$



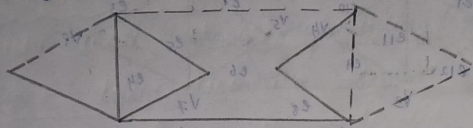
3. Choose the edge $e_3 = v_6 v_8$ then $w_4 = \{v_1 e_1 v_2 e_2 e_3 v_8\}$



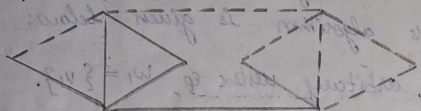
4. Choose the edge $e_4 = v_8 v_7$ then $w_5 = \{v_1 e_1 v_2 e_2 v_6 e_3 v_8 e_4 v_7\}$



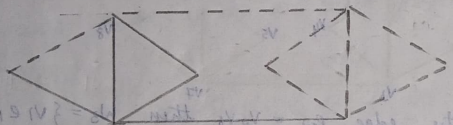
5. Choose $e_5 = v_7 v_6$ then $w_6 = \{v_1 e_1 v_2 e_2 v_3 e_3 v_8 e_4 v_7 e_5 v_6\}$



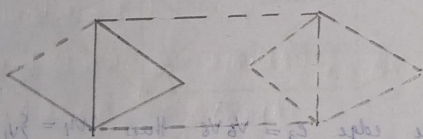
6. Choose $e_7 = v_6 v_5$ then $w_7 = \{v_1 e_1 v_2 e_2 v_6 e_3 v_4 e_4 v_7 e_5 v_6 e_7 v_5\}$



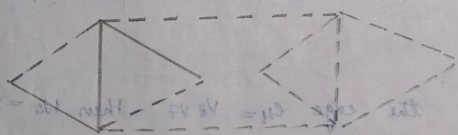
7. Choose $e_6 = v_5 v_7$ then $w_8 = \{v_1 e_1 v_2 e_2 v_6 e_3 v_8 e_4 v_7 e_6 v_5\}$



8. Choose $e_8 = v_7 v_3$ then $w_9 = \{v_1 e_1 v_2 e_2 v_6 e_3 v_8 e_4 v_7 e_8 v_3\}$

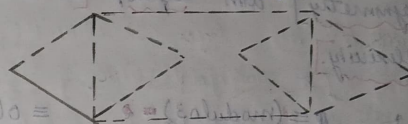


9. Choose $e_9 = v_3 v_4$ then $w_{10} = \{v_1 e_1 v_2 e_2 v_6 e_3 v_8 e_4 v_7 e_5 v_6 e_9 v_4\}$

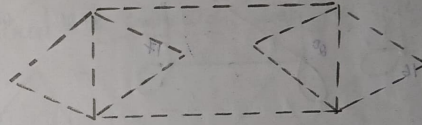


10. Choose $e_{10} = v_4 v_2$ then $w_{11} = \{v_1 e_1 v_2 e_{10} v_6 e_3 v_8 e_4 v_7 e_5 v_6 e_9 v_4\}$

11. Now take $e_{11} = v_2 v_3$ then $w_{12} = \{v_1 e_1 v_2 e_{11} v_3 e_2 v_6 e_3 v_8 e_4 v_7 e_5 v_6 e_9 v_4 e_{10} v_2\}$



12. Now take $e_{12} = v_3 v_1$ then $w_{13} = \{v_1 e_{12} v_2 e_2 v_6 e_3 v_8 e_4 v_7 e_5 v_6 e_9 v_4 e_{10} v_2 e_{11} v_3 e_{12} v_1\}$



3/2/15
Monday Equivalence relation:

A binary relation is called an equivalence relation if it is reflexive, symmetric & transitive.

Eg: '=' (is equal to), '||' (parallel to), 'is isomorphic to'

A graph representing an equivalence relation is called a equivalence graph.

Q. Draw Equivalence graph for binary relation "=" mod 3 on $\{10, \dots, 20\}$.
Ans: imp Equivalence graph for the relation is "congruent to modulo 3." defined on a set of 11 integers, 10 through 20 is given below. i.e the $\{10, 11, 12, \dots, 20\}$

Here each undirected edges represent parallel but

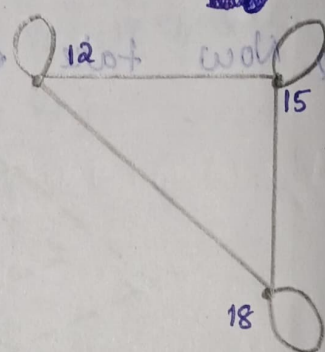
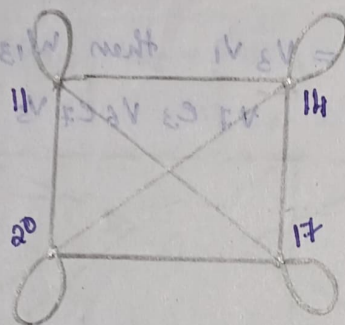
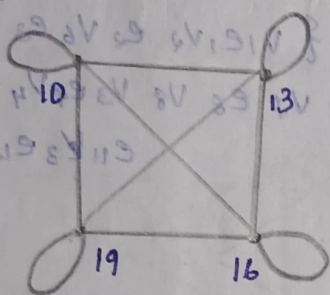
opposite directed edges.

Here vertex set of graph is divided into 3 disjoint classes each in separate components. Each component is an undirected subgraph [due to symmetry] with selfloop [at each vertex due to reflexivity].

$$\equiv 1 \pmod{3}$$

$$\equiv 2 \pmod{3}$$

$$\equiv 0 \pmod{3}$$



Note:

$$\left. \begin{array}{l} 10 \pmod{3} = 1 \\ 13 \pmod{3} = 1 \end{array} \right\} \equiv 1 \pmod{3}$$

$$14 \pmod{3} = 2 \equiv 2 \pmod{3}$$

$$12 \pmod{3} = 0 \equiv 0 \pmod{3}$$

Equivalence classes are $\{10, 13, 16, 19\}$ $\{11, 14, 17, 20\}$

$\{12, 15, 18\}$

Reflexive $\rightarrow$ loop at every node.

Symmetric $\rightarrow u \rightarrow v, v \rightarrow u$